Memory Management & Address Translation

Going Beyond Physical Memory

- **Swap Space**
  - Reserved area on disk used to store pages that are not currently in RAM
    - Acts as overflow when physical memory is insufficient
    - Allows the system to maintain larger logical address spaces than physical RAM alone
- **Page Fault**
  - Event triggered when a program accesses a page not present in RAM
    - Hardware signals the fault and the OS must fetch the page from swap into RAM
    - Handling a page fault involves loading the page and resuming the faulting access
- **Page Replacement Policies**
  - Mechanisms for choosing which page to evict when RAM is full
    - FIFO (First-In, First-Out)
    - LRU (Least Recently Used), which evicts pages not touched for the longest time
- **Thrashing**
  - Condition where the system spends more time swapping pages than executing code
    - Occurs when the combined working sets of processes exceed available physical RAM
    - Leads to severe performance degradation and excessive disk I/O

Virtual Memory: Paging

- **Paging**
  - Memory is divided into fixed-size units called pages (virtual) and frames (physical)
    - Fixed-size units eliminate external fragmentation
    - Allows noncontiguous physical memory allocation for a process
- **Page Tables**
  - OS data structure that maps a process's virtual pages to physical frames
    - Stores the mapping information required for address translation
    - Lookups may require accessing memory, which can be slow without caching
- **TLBs (Translation Lookaside Buffers)**
  - High-speed hardware cache of recently used address translations
    - TLB hit returns translation quickly (fast)
    - TLB miss requires a page table lookup in RAM (slow)
- **Multi-level Page Tables**
  - Technique that reduces page table memory by creating mapping structures only for used parts of the address space
    - Breaks the page table into hierarchical levels so unused regions do not consume memory
    - Useful for sparse address spaces

Address Space and Address Translation

- **Address Space**
  - OS abstraction that makes each process appear to have its own private block of memory
    - This logical address space contains the program's code, stack, and heap
    - Provides isolation so processes do not directly access each other's physical memory
- **Address Translation**
  - Hardware mechanism that converts virtual addresses to physical addresses
    - The CPU performs the transformation so programs use virtual addresses while RAM uses physical addresses
    - Ensures the illusion of a contiguous private address space for each process
- **Dynamic Relocation (Base and Bounds)**
  - Simple hardware relocation using two registers per CPU: Base and Bounds
    - Base holds the start of the process's physical memory block
    - Bounds holds the size of that block to enforce limits
  - Physical address calculation uses the base register
    - Physical Address = Virtual Address + Base
- **Segmentation**
  - Memory divided into logical segments such as code, stack, and heap
    - Reduces internal fragmentation compared to a single contiguous block
    - Aligns memory layout with program structure and protection needs
  - Enables sharing of segments between processes
    - Example: multiple processes can share the same read-only code segment

Presented with xmind

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